

OpenAI Agents SDK: From Zero to Your First Tool-Using Agent

A hands-on guide for beginners

This guide takes you from nothing to a working AI agent, in the order you should actually learn it:

1. **Setup** — get a project ready.
2. **OpenAI key integration** — the simplest possible agent (SDK's native, default provider).
3. **Gemini key integration** — swap in a different LLM provider.
4. **Runner and asyncio** — how an agent actually executes, and why it's async under the hood.
5. **Model configuration** — the three levels (Agent / Run / Global) for choosing which LLM to use.
6. **Basic tool use** — give your agent the ability to call Python functions with `@function_tool`.
7. **Model settings** — turn the model's knobs: temperature, tool choice, answer length.
8. **Local context** — hand your tools data about the user that the LLM never sees.
9. **Dynamic instructions** — build the system prompt at runtime instead of hardcoding it.
10. **Agent cloning** — spin specialised variants off one base agent.
11. **Tracing** — watch every step of a run on the traces dashboard.
12. **Agents as tools** — let one agent call another as a specialist.
13. **Handoffs** — transfer the whole conversation to a specialist agent.
14. **Advanced tool control** — stop the loop, gate tools, survive failures.

Everything here is copy-paste ready. Type the code yourself rather than pasting — it sticks better.

Part 0 — Setup

You need **Python 3.13+** and **uv** (a fast Python package manager). If you don't have **uv**:

```
pip install uv
```

Create a project:

```
uv init hello-agent
cd hello-agent
uv add openai-agents python-dotenv
```

Create a `.env` file in the project root — this is where your API keys live. **Never commit this file.**

```
OPENAI_API_KEY=
GEMINI_API_KEY=
```

You'll fill these in as you go through Parts 1 and 2.

Part 1 — OpenAI Key Integration (the default provider)

The Agents SDK is built by OpenAI, so **OpenAI is the default provider** — no custom client setup required. You just need an API key.

Get a key

1. Go to <https://platform.openai.com/api-keys> and create a key.
2. Paste it into `.env`:

```
OPENAI_API_KEY=sk-...
```

Write the agent

Create `main.py`:

```
from agents import Agent, Runner
from dotenv import load_dotenv, find_dotenv

# Loads .env so OPENAI_API_KEY is available as an environment variable
load_dotenv(find_dotenv())

# The SDK finds OPENAI_API_KEY automatically — no client wiring needed
agent = Agent(
    name="Assistant",
    instructions="You are a helpful assistant.",
)

result = Runner.run_sync(agent, "Write a haiku about recursion in programming.")
print(result.final_output)
```

Run it:

```
uv run main.py
```

That's the whole integration. Three things happened:

- `Agent(...)` — defined *who* the agent is (a name + a system prompt via `instructions`).
- `Runner.run_sync(...)` — actually sent the prompt to the LLM and waited for the result.
- `result.final_output` — the agent's final text response.

Note: If you don't pass a `model=`, the Agent uses the SDK's built-in default OpenAI model. You'll learn to override this explicitly in Part 4.

Part 2 — Gemini Key Integration (a different provider)

The Agents SDK only *ships* with OpenAI wiring, but it can talk to **any provider that exposes an OpenAI-compatible Chat Completions API** — and Google's Gemini API does exactly that.

Get a key

1. Go to <https://ai.google.dev/gemini-api/docs/api-key> and create a Gemini API key.
2. Paste it into `.env`:

```
GEMINI_API_KEY=...
```

Write the agent

```
from agents import Agent, Runner, OpenAIChatCompletionsModel, AsyncOpenAI
import os
from dotenv import load_dotenv, find_dotenv

load_dotenv(find_dotenv())

# 1. Point an OpenAI-compatible client at Gemini's endpoint instead of OpenAI's
external_client = AsyncOpenAI(
    api_key=os.getenv("GEMINI_API_KEY"),
    base_url="https://generativelanguage.googleapis.com/v1beta/openai/",
)

# 2. Wrap it as a Chat Completions model the SDK understands
llm_model = OpenAIChatCompletionsModel(
    model="gemini-2.5-flash",
    openai_client=external_client,
)

# 3. Hand that model to the Agent instead of using the default
agent = Agent(name="Assistant", model=llm_model)

result = Runner.run_sync(agent, "Welcome and motivate me to learn Agentic AI.")
print("AGENT RESPONSE:", result.final_output)
```

Why this works

- `AsyncOpenAI` is just an HTTP client shaped for OpenAI's API contract. Gemini implements that same contract at a different `base_url`, so the same client class works — you're not using any Google-specific SDK.
- `OpenAIChatCompletionsModel` tells the Agent "call this client using the Chat Completions format, with this specific model name."

- Everything else — `Agent`, `Runner`, `.final_output` — is identical to Part 1. **This is the whole point of the SDK's design:** the provider is just a pluggable `model`, the rest of your agent code never changes.

Part 3 — Understanding `Runner` and `asyncio`

You've used `Runner.run_sync(...)` twice without asking what it does. Time to open that up.

Why agents are async in the first place

Every call an agent makes to an LLM is a **network request** — it goes out over HTTP and the program sits waiting for a response. That's *I/O-bound* waiting, not CPU work. Python's `asyncio` exists so a program can do other useful things (or run several LLM calls concurrently) instead of blocking dead while it waits on the network. The Agents SDK is written natively with `async/await` for this reason.

The three ways to run an agent

Method	Signature	When to use it
<code>Runner.run(...)</code>	<code>async def run(...)</code> — must be <code>awaited</code>	Inside an <code>async def</code> function — real applications, servers, concurrent agents
<code>Runner.run_sync(...)</code>	plain function	Scripts, notebooks, quick tests — internally starts an event loop and blocks for you
<code>Runner.run_streamed(...)</code>	returns a streaming result	When you want to show tokens/events as they arrive, instead of waiting for the whole answer

The `async` version, properly

```
import asyncio
from agents import Agent, Runner

agent = Agent(name="Assistant", instructions="You only respond in haikus.")

async def main():
    result = await Runner.run(agent, "Tell me about recursion in programming.")
    print(result.final_output)

if __name__ == "__main__":
    asyncio.run(main())
```

Breaking this down:

- `async def main()` — declares a coroutine, a function that can pause and resume instead of blocking.
- `await Runner.run(...)` — pause `main()` here until the LLM responds, without freezing the whole program.

- `asyncio.run(main())` — creates the event loop, runs `main()` to completion, and shuts the loop down. This is the standard entry point for any async Python script.

`Runner.run_sync(agent, "...")` you used in Parts 1–2 is a convenience wrapper that does exactly `asyncio.run(Runner.run(agent, "..."))` under the hood. **Use `run_sync` for quick scripts. Use `run` inside real async apps** — e.g. a FastAPI endpoint, or when you want to fire off multiple agents at once with `asyncio.gather(...)`.

Try it: run two agents concurrently

```
import asyncio
from agents import Agent, Runner

agent = Agent(name="Assistant", instructions="Answer in one short sentence.")

async def main():
    results = await asyncio.gather(
        Runner.run(agent, "What is Python?"),
        Runner.run(agent, "What is an AI agent?"),
    )
    for r in results:
        print(r.final_output)

asyncio.run(main())
```

Both prompts are in flight to the LLM provider at the same time — this is only possible because `Runner.run` is async.

Part 4 — Model Configuration: Agent, Run, and Global Levels

You already did **Agent-level** configuration in Part 2 (passing `model=` straight into `Agent(...)`). That's the SDK's default provider is OpenAI; you can override it at three different scopes.

We will always use Agent-level configuration in this course — it lets each agent in a multi-agent system use whichever model fits its job (e.g., a cheap fast model for routing, a stronger model for reasoning).

1. Agent Level — override per agent

```
from agents import Agent, OpenAIChatCompletionsModel, AsyncOpenAI, Runner, set_tracing_disabled

client = AsyncOpenAI(
    api_key=gemini_api_key,
    base_url="https://generativelanguage.googleapis.com/v1beta/openai/",
)
set_tracing_disabled(disabled=True) # OpenAI's tracing dashboard expects an OpenAI key; disable

agent = Agent(
    name="Assistant",
    instructions="You only respond in haikus.",
    model=OpenAIChatCompletionsModel(model="gemini-2.0-flash", openai_client=client),
)
```

Different agents in the same app can each have their own `model=`.

2. Run Level — override for one call

```
from agents.run import RunConfig

config = RunConfig(
    model=OpenAIChatCompletionsModel(model="gemini-2.0-flash", openai_client=external_client),
    model_provider=external_client,
    tracing_disabled=True,
)

agent = Agent(name="Assistant", instructions="You are a helpful assistant")
result = Runner.run_sync(agent, "Hello, how are you.", run_config=config)
```

The agent itself is unconfigured; the override applies only to this specific `Runner.run_sync(...)` call — useful for A/B testing a model without touching agent definitions.

3. Global Level — override for the whole process

```
from agents import set_default_openai_client, set_default_openai_api, set_tracing_disabled

set_tracing_disabled(True)
set_default_openai_api("chat_completions")
set_default_openai_client(external_client)

agent = Agent(name="Assistant", instructions="You are a helpful assistant", model="gemini-2.0-")
```

Every agent created afterward uses this client by default, unless it — or its run — overrides it locally. Agent-level and Run-level configuration both win over Global.

Precedence, most specific wins: Agent level > Run level > Global level.

Part 5 — Basic Tool Use with `@function_tool`

So far your agent can only talk. A **tool** lets it *do* something — call a Python function, hit an API, query a database — and use the result to answer better.

The core idea

You write a normal Python function. The `@function_tool` decorator turns it into something the LLM can call:

- The **function name** becomes the tool's name.
- The **docstring** becomes the tool's description (this is what tells the LLM *when* to use it).
- The **type hints** are used to auto-generate the JSON schema for the tool's parameters.

```
from agents import Agent, Runner, function_tool

@function_tool
def get_weather(city: str) -> str:
    """Get the current weather for a given city."""
    return f"The weather in {city} is sunny and 25°C."

agent = Agent(
    name="Weather Assistant",
    instructions="You are a helpful assistant. Use tools when they help you answer.",
    tools=[get_weather],
)

result = Runner.run_sync(agent, "What's the weather like in Karachi?")
print(result.final_output)
```

What actually happens when you run this

1. `Runner.run_sync` sends your prompt **and** the tool's schema (name, description, parameters) to the LLM.
2. The LLM decides the question needs live data it doesn't have, and responds with "call `get_weather` with `city="Karachi"`" instead of a text answer.
3. The SDK intercepts that, **runs your actual Python function** with those arguments, and sends the return value back to the LLM as a new message.
4. The LLM reads the tool's result and writes the final natural-language answer.
5. `Runner.run_sync` only returns to you once this whole loop is finished — `result.final_output` is the final answer, tool calls included.

You can register multiple tools; the agent decides on its own which (if any) to call, and can even chain several tool calls in one turn.

Try it: add a second tool

```
@function_tool
def add(a: int, b: int) -> int:
    """Add two numbers together."""
    return a + b

agent = Agent(
    name="Assistant",
    instructions="Use tools when they help you answer.",
    tools=[get_weather, add],
)

result = Runner.run_sync(agent, "What's 42 plus 58, and what's the weather in Lahore?")
print(result.final_output)
```

Watch the model correctly call both tools in the same turn and combine both results into one answer.

Beyond `@function_tool`

`@function_tool` is for **your own** Python functions. The SDK also ships **hosted/built-in OpenAI tools** — like `WebSearchTool` and `FileSearchTool` — that run on OpenAI's servers instead of your machine. Those are out of scope for this guide; `@function_tool` is the one you'll use 90% of the time, and it's the foundation everything else builds on.

Part 6 — Model Settings: Temperature, Tool Choice, and Length

Part 4 decided **which** model runs your agent. `ModelSettings` decides **how** that model behaves once it's running — how creative it is, whether it's allowed to reach for a tool, and how long its answer can get. Same agent, same prompt, different output.

The three settings to learn first

- **temperature** — the creativity dial. Low keeps answers focused and repeatable; high makes them varied and surprising.
- **tool_choice** — whether the model may, must, or must not call a tool.
- **max_tokens** — a hard ceiling on how long the answer can be.

You pass them as one object to the Agent:

```

from agents import Agent, Runner, ModelSettings

question = "Tell me about AI in 2 sentences."

for temperature in (0.1, 1.9):
    agent = Agent(
        name="Assistant",
        instructions="You are a helpful assistant.",
        model=llm_model,
        model_settings=ModelSettings(temperature=temperature),
    )
    print(f"\ntemperature={temperature}:")
    print(Runner.run_sync(agent, question).final_output)

```

Run it twice. The **0.1** answer will barely change between runs; the **1.9** answer will come out different every time. Use low temperature for maths, facts and instruction-following; high for brainstorming and story writing.

Note: The usable range depends on the provider. Gemini accepts a **temperature** up to **2.0**, so **1.9** is a legal (and very loose) setting here.

Forcing — or forbidding — tool use

tool_choice overrides the model's own judgement about the tools you registered in Part 5.

```

from agents import function_tool

@function_tool
def calculate_area(length: float, width: float) -> str:
    """Calculate the area of a rectangle."""
    return f"Area = {length} × {width} = {length * width} square units"

question = "What's the area of a 5x3 rectangle?"

for choice in ("auto", "required", "none"):
    agent = Agent(
        name=f"{choice} agent",
        instructions="You are a helpful assistant.",
        model=llm_model,
        tools=[calculate_area],
        model_settings=ModelSettings(tool_choice=choice),
    )
    print(f"\ntool_choice={choice}:")
    print(Runner.run_sync(agent, question).final_output)

```

- **"auto"** — the model decides. This is the default, and what Part 5 was already doing.
- **"required"** — the model **must** call a tool before it may answer. Useful when guessing is worse than failing (calculations, lookups).

- **"none"** — the model still sees the tool's schema but is forbidden from calling it. It will answer from its own head, and you'll see it do the arithmetic itself.

Try it: cap the answer length

```
brief_agent = Agent(  
    name="Brief Assistant",  
    instructions="You are a helpful assistant.",  
    model=llm_model,  
    model_settings=ModelSettings(temperature=0.2, max_tokens=100),  
)
```

max_tokens cuts the response off at the limit — it does **not** ask the model to be concise. If the answer needs more room than you gave it, it stops mid-sentence. Say "answer in two sentences" in the **instructions** when you want brevity; use **max_tokens** when you want a cost ceiling.

The rest of the knobs

ModelSettings also carries **parallel_tool_calls** (let the agent fire several tools at once instead of one at a time), **top_p**, **frequency_penalty** and **presence_penalty**. Leave all of them at their defaults until you have a specific reason not to — changing one setting at a time is the only way to tell what actually helped.

Part 7 — Local Context: Data Your Tools See and the LLM Doesn't

Your tools so far have been self-contained: everything they needed arrived as an argument from the LLM. Real tools need things the LLM shouldn't be inventing — who the logged-in user is, their ID, a database handle, a logger. **Local context** is how you pass those in.

The core idea

You define a plain Python object (a dataclass is the usual choice), hand it to **Runner.run(..., context=...)**, and the SDK wraps it in a **RunContextWrapper** that every tool in that run receives as its first parameter.

```

import asyncio
from dataclasses import dataclass
from agents import Agent, Runner, RunContextWrapper, function_tool

@dataclass
class UserInfo:
    name: str
    uid: int

@function_tool
async def fetch_user_age(wrapper: RunContextWrapper[UserInfo]) -> str:
    """Returns the age of the user."""
    return f"User {wrapper.context.name} is 47 years old"

async def main():
    user_info = UserInfo(name="Ali", uid=123)

    agent = Agent[UserInfo](
        name="Assistant",
        model=llm_model,
        tools=[fetch_user_age],
    )

    result = await Runner.run(
        starting_agent=agent,
        input="What is the age of the user?",
        context=user_info,
    )
    print(result.final_output)

asyncio.run(main())

```

Note what the prompt does **not** say: it never mentions Ali. The LLM asks for the user's age, the SDK runs your function, and your function reads the name out of `wrapper.context`.

The part that surprises people

The context is **never sent to the LLM**. Look at the schema the SDK generated for that tool:

```

print(fetch_user_age.params_json_schema)
# {'properties': {}, 'title': 'fetch_user_age_args', 'type': 'object', ...}

```

Empty. The `wrapper` parameter is stripped out before the tool description goes to the model, so the model calls `fetch_user_age()` with no arguments and can't see — or invent — the user's identity. Compare that with `calculate_area` in Part 6, where `length` and `width` are in the schema because the model is meant to supply them.

Two rules follow from this:

- Every tool, hook and callback in a single run must use the **same** context type — that's why the agent is written `Agent[UserInfo]`.
- If you want the LLM to *know* something, it belongs in `instructions` or in the input message, not in the context.

Try it: a second tool on the same context

```
@dataclass
class UserInfo:
    name: str
    uid: int
    location: str = "Pakistan"

@function_tool
async def fetch_user_location(wrapper: RunContextWrapper[UserInfo]) -> str:
    """Returns the location of the user."""
    return f"User {wrapper.context.name} is from {wrapper.context.location}"
```

Register both tools, then ask "What is the age of the user, and where are they from?" — one context object, two tools reading different fields out of it, no user data anywhere in the prompt.

Part 8 — Dynamic Instructions: A System Prompt That Changes

Every agent so far had its `instructions` fixed at definition time. The SDK also accepts a `function` there. It runs at the start of each turn and returns the string to use as that turn's system prompt — so the agent's persona can depend on who is asking, what time it is, or how far the conversation has gone.

The core idea

The function takes exactly two parameters — the context wrapper from Part 7 and the agent itself — and returns a `str`.

```

from dataclasses import dataclass
from agents import Agent, Runner, RunContextWrapper

@dataclass
class UserInfo:
    name: str
    uid: int

def special_prompt(ctx: RunContextWrapper[UserInfo], agent: Agent) -> str:
    return (
        f"You are a math expert. User: {ctx.context.name}, Agent: {agent.name}. "
        "Please assist with math-related queries."
    )

math_agent = Agent[UserInfo](
    name="Genius",
    instructions=special_prompt,
    model=llm_model,
)

user = UserInfo(name="Ali", uid=123)
result = Runner.run_sync(math_agent, "What is 17 × 23?", context=user)
print(result.final_output)

```

- `ctx.context` is the object you passed as `context=` in Part 7 — this is where local context and instructions meet.
- `agent` is the agent being run, so `agent.name` and `len(agent.tools)` are available while you build the prompt.
- The function may also be `async def`; the SDK awaits it, so you can hit a database before returning the string.

Note: The callable must accept **exactly two** parameters. Give it one or three and the SDK raises `TypeError: 'instructions' callable must accept exactly 2 arguments (context, agent)`.

Try it: instructions that remember

Any callable object works, not just a function — which is how you keep state between runs:

```

class StatefulInstructions:
    def __init__(self):
        self.count = 0

    def __call__(self, ctx: RunContextWrapper, agent: Agent) -> str:
        self.count += 1
        if self.count == 1:
            return "You are a learning assistant. First interaction – be welcoming."
        return f"You are a learning assistant. Interaction #{self.count} – be brief."

agent = Agent(name="Stateful", instructions=StatefulInstructions(), model=llm_model)

for _ in range(3):
    print(Runner.run_sync(agent, "Tell me about AI in one line").final_output)

```

The first answer greets you; the next two get progressively terser, because the object counted the runs.

What the context does not hold

`RunContextWrapper` carries `.context` (your own object), `.usage` (token counts so far) and `.turn_input`. It does **not** carry a list of past messages. Code like `len(getattr(ctx, "messages", []))` compiles, runs, and quietly returns `0` forever — the branch you wrote for a long conversation never fires. If you want to react to conversation length, count it yourself in your own context object, the way `StatefulInstructions` above does.

Part 9 — Cloning Agents: One Base, Many Variants

Parts 6 and 8 handed you knobs: temperature, tool choice, instructions. Once you have an agent you like, you rarely want to rewrite it from scratch just to turn one of those. `agent.clone(...)` copies an agent and overrides only the fields you name.

The core idea

```
from agents import Agent, Runner, ModelSettings

base_agent = Agent(
    name="BaseAssistant",
    instructions="You are a helpful assistant.",
    model=llm_model,
    model_settings=ModelSettings(temperature=0.7),
)

creative_agent = base_agent.clone(
    name="CreativeAssistant",
    instructions="You are a creative writing assistant. Use vivid language.",
    model_settings=ModelSettings(temperature=0.9),
)

precise_agent = base_agent.clone(
    name="PreciseAssistant",
    instructions="You are a precise, factual assistant.",
    model_settings=ModelSettings(temperature=0.1),
)

for agent in (base_agent, creative_agent, precise_agent):
    print(f"\n{agent.name}:")
    print(Runner.run_sync(agent, "Describe a sunset.").final_output)
```

Anything you don't pass is carried over from the base — neither clone repeats `model=llm_model`, and both still run on Gemini. This is how you build a routing model, a reasoning model and a summariser out of one definition.

The shallow-copy trap

`clone()` is `dataclasses.replace` underneath, which makes a **shallow** copy. A list you don't override isn't copied — the base and the clone hold the *same* list object.

```
base = Agent(name="Base", instructions="Be helpful.", tools=[get_weather])
shared_clone = base.clone(name="SharedClone")

base.tools.append(add)          # appended through the original...
print(len(shared_clone.tools))  # 2 — the clone grew a tool too

independent = base.clone(name="Independent", tools=[*base.tools])
base.tools.append(calculate_area)
print(len(independent.tools))   # 2 — its own list, unaffected
```

The rule: **pass a fresh list whenever you want independence** — `tools=[*base.tools, extra_tool]`. The same applies to `handoffs`, `mcp_servers` and the guardrail lists.

Try it: settings are replaced, not merged

```
from dataclasses import replace

base = Agent(
    name="Base",
    instructions="Be helpful.",
    model_settings=ModelSettings(temperature=0.7, max_tokens=500),
)

careless = base.clone(model_settings=ModelSettings(temperature=0.1))
print(careless.model_settings.max_tokens)    # None – the 500 was dropped

careful = base.clone(model_settings=replace(base.model_settings, temperature=0.1))
print(careful.model_settings.max_tokens)     # 500 – every other field kept
```

A `ModelSettings` handed to `clone()` swaps the whole object, so any field you leave out silently falls back to its default. When you mean to change one setting and keep the rest, `dataclasses.replace` on the base's settings is the safe move.

Part 10 — Tracing: Seeing What Your Agent Actually Did

Since Part 4 you've been writing `set_tracing_disabled(disabled=True)` without being told what you were switching off. This is it. Tracing is on by default, and it records every step of a run so you can open it up afterwards instead of guessing from the final answer.

Trace and span

A **trace** is one complete workflow — everything that happened between your prompt and the final output. A **span** is one step inside it: an LLM call, a tool call, one agent's turn. A run that calls `get_weather` and then answers produces one trace containing several spans, and the dashboard draws them as a timeline.

Turning it on while using Gemini

Here's the wrinkle that trips everyone up: your **model** is Gemini, but the **trace dashboard** is OpenAI's. Exporting traces needs an OpenAI key even though inference never touches OpenAI.

```

import os
from dotenv import load_dotenv, find_dotenv
from agents import Agent, Runner, function_tool, set_tracing_export_api_key

load_dotenv(find_dotenv())

# Inference runs on Gemini; this key is only used to upload traces
set_tracing_export_api_key(os.getenv("OPENAI_API_KEY", ""))

@function_tool
def get_weather(city: str) -> str:
    """A simple function to get the weather for a user."""
    return f"The weather for {city} is sunny."

agent = Agent(
    name="WeatherAgent",
    instructions="You are a helpful assistant.",
    model=llm_model,
    tools=[get_weather],
)

result = Runner.run_sync(agent, "What's the weather in Karachi?")
print(result.final_output)

```

Note what is **missing**: no `set_tracing_disabled(...)`. Run it, then open <https://platform.openai.com/traces> and you'll see the run, the tool call, and the model's second pass over the tool result.

Note: If the key is missing the SDK doesn't crash — it prints `OPENAI_API_KEY is not set, skipping trace export` and carries on. An empty dashboard usually means that line scrolled past you.

Try it: group several runs into one trace

By default every `Runner.run(...)` is its own trace. Wrap them to get one story instead of three:

```

import asyncio
from agents import trace

async def main():
    with trace("Joke workflow"):
        first = await Runner.run(agent, "Tell me a joke")
        second = await Runner.run(agent, f"Rate this joke: {first.final_output}")
        print(f"Joke: {first.final_output}")
        print(f"Rating: {second.final_output}")

asyncio.run(main())

```

Both runs now appear as spans under a single **Joke workflow** trace, in order, with timings.

Turning it back off

`set_tracing_disabled(True)` kills tracing for the whole process; `RunConfig(tracing_disabled=True)` kills it for one run. If you want the timeline but not the content, `RunConfig(trace_include_sensitive_data=False)` keeps prompts and outputs out of the upload.

Part 11 — Agents as Tools: One Agent Calling Another

In Part 5 a tool was a Python function. A tool can also be **another agent**. You keep one agent in charge of the conversation and let it call specialists the way it calls `get_weather` — each with its own tight instructions, none of them taking over the conversation.

The core idea

`as_tool()` wraps any agent so another agent can call it:

```

from agents import Agent, Runner

spanish = Agent(
    name="Spanish Translator",
    instructions="Translate what the user says into Spanish. Only output Spanish.",
    model=llm_model,
)

summarizer = Agent(
    name="Summarizer",
    instructions="Summarize the given text in 2 short bullet points.",
    model=llm_model,
)

coach = Agent(
    name="Writing Coach",
    instructions=(
        "You help users improve messages.\n"
        "- If they ask for Spanish, call translate_to_spanish.\n"
        "- If they ask for a summary, call summarize_text.\n"
        "- Otherwise, give a short tip yourself."
    ),
    model=llm_model,
    tools=[
        spanish.as_tool(
            tool_name="translate_to_spanish",
            tool_description="Translate user text to Spanish.",
        ),
        summarizer.as_tool(
            tool_name="summarize_text",
            tool_description="Summarize text in 2 bullets.",
        ),
    ],
)

result = Runner.run_sync(coach, "Please translate to Spanish: I love hands-on examples.")
print(result.final_output)

```

The reply comes from the **coach**, not from the translator. The specialist ran, returned its text, and the coach wrote the final answer around it — exactly the loop from Part 5, with an agent where the Python function used to be.

- The generated tool takes a single string parameter, `input`. The orchestrator writes that string itself, so its `instructions` decide what the specialist actually receives.
- `tool_description` is what the orchestrator reads when choosing — same rule as the docstring in Part 5.
- Each specialist keeps its own `model` and `model_settings`, so a cheap fast model can do translation while the orchestrator runs on a stronger one.

Try it: full control with a function tool

When you need more than the wrapper offers — a turn limit, post-processing, a different run config — write an ordinary `@function_tool` and call `Runner.run` inside it:

```
from agents import function_tool

proofreader = Agent(
    name="Proofreader",
    instructions="Fix grammar and punctuation. Reply only with the corrected text.",
    model=llm_model,
)

@function_tool
async def proofread_text(text: str) -> str:
    """Fix grammar and punctuation; return only the corrected text."""
    result = await Runner.run(proofreader, text, max_turns=3)
    return str(result.final_output)
```

Agent-as-tool is not a handoff

A **handoff** transfers the conversation: the second agent takes the mic and answers the user directly. **Agent-as-tool** keeps the first agent in charge — it borrows an answer, then replies in its own voice. Use tools when you want one consistent personality with borrowed skills; handoffs are for genuinely passing the job on — which is exactly what Part 12 does.

Part 12 — Handoffs: Transferring the Conversation

Part 11 ended with the distinction: a tool borrows an answer, a **handoff** passes the job on. This is the handoff. The first agent stops being the one talking to the user, and a specialist takes over the rest of the conversation.

The core idea

List the agents an agent may transfer to in `handoffs=`. The SDK turns each one into a tool the model can call — a **Study Coach** becomes `transfer_to_study_coach` — so routing is just tool selection, using everything you already know from Part 5.

```

import asyncio
from agents import Agent, Runner, handoff

fitness_coach = Agent(
    name="Fitness Coach",
    instructions=(
        "You're a running coach. Ask 1-2 quick questions, then give a week plan. "
        "Keep it simple and encouraging. No medical advice."
    ),
    model=llm_model,
)

study_coach = Agent(
    name="Study Coach",
    instructions=(
        "You're a study planner. Ask for the current routine, then give a "
        "one-week schedule. Keep steps small and doable."
    ),
    model=llm_model,
)

router = Agent(
    name="Coach Router",
    instructions=(
        "Route the user:\n"
        "- running, workout, stamina -> hand off to the Fitness Coach.\n"
        "- exams, study plan, focus, notes -> hand off to the Study Coach.\n"
        "After the handoff the specialist continues the conversation."
    ),
    model=llm_model,
    handoffs=[study_coach, handoff(fitness_coach)],
)

async def main():
    result = await Runner.run(router, "I want to run 5km in 8 weeks. Can you help?")
    print(result.final_output)
    print("answered by:", result.last_agent.name)

asyncio.run(main())

```

Both forms in that list are legal: a bare agent, or `handoff(agent)` when you want to customise it. The router's own reply never reaches the user — the coach's does.

Proving the handoff happened

`result.last_agent` is the agent that actually answered, and it's how you know routing worked. For the full evidence, look through `result.new_items` for a `HandoffCallItem` (the router asked) followed by a `HandoffOutputItem` (the specialist accepted).

Note: The generated tool is named from the agent's name, lowercased with underscores — `Fitness Coach` becomes `transfer_to_fitness_coach`. Rename an agent and you rename its handoff tool, which quietly invalidates any instructions that referred to the old name.

Try it: keep talking to the specialist

A handoff transfers the conversation, so the *next* turn should go straight to the specialist rather than back through the router. Back inside `main()`, after that first run:

```
follow_up = result.to_input_list() + [
    {"role": "user", "content": "Right now I jog about 2 km, 3 days a week."}
]
second = await Runner.run(result.last_agent, follow_up)
print(second.final_output)
```

`to_input_list()` hands you the whole conversation so far in the format `Runner.run` expects. Pair it with `result.last_agent` and the coach picks up mid-thought instead of reintroducing itself.

Customising a handoff

`handoff(agent, ...)` takes `tool_name_override` and `tool_description_override` when the generated names read badly to the model, `on_handoff` for a callback that fires at the moment of transfer (logging, warming a cache), `input_type` to demand structured data from the router, and `input_filter` to trim what history the specialist inherits. Start with bare agents; reach for these when routing misbehaves.

Part 13 — Advanced Tool Control: Stopping, Gating, and Failing Well

Part 5 gave your agent tools. This part is about control: when the loop stops, which tools exist for which user, and what happens when a tool blows up. These are the settings that separate a demo from something you'd let other people run.

Stopping the loop early

By default a tool's output goes back to the LLM, which decides what to do next — that's `tool_use_behavior="run_llm_again"`. Two other modes cut that short. `"stop_on_first_tool"` makes the first tool's raw output the final answer. `StopAtTools` names the tools that end the workflow:

```

from agents import Agent, Runner, function_tool, StopAtTools

@function_tool
def get_weather(city: str) -> str:
    """A simple function to get the weather for a user."""
    return "Sunny"

@function_tool
def get_travel_plan(city: str) -> str:
    """Plan travel for the given city."""
    return "Travel plan is not available"

agent = Agent(
    name="TravelAgent",
    instructions="You are a helpful assistant.",
    model=llm_model,
    tools=[get_weather, get_travel_plan],
    tool_use_behavior=StopAtTools(stop_at_tool_names=["get_travel_plan"]),
)

result = Runner.run_sync(agent, "Make me a travel plan for Lahore")
print(result.final_output)

```

`final_output` is now the tool's own string, word for word. The model never saw it, never softened it, never added a sentence around it. That's the point when a tool already returns exactly what the user should get — and the trap when it doesn't.

The safety net: `max_turns`

`max_turns` caps how many times the loop may call the LLM. Reach it and the SDK raises, rather than burning your quota in a circle:

```

from agents.exceptions import MaxTurnsExceeded

try:
    result = Runner.run_sync(agent, "Research AI agents thoroughly.", max_turns=3)
    print(result.final_output)
except MaxTurnsExceeded:
    print("Hit the turn limit – stopping instead of looping forever.")

```

Count the turns your workflow actually needs before setting this. A tool call plus the model's reading of the result is two turns, so `max_turns=1` stops before the agent can ever use a tool.

Tools that appear and disappear

`is_enabled` decides whether a tool is offered to the model at all. Pass `False` to shelve a tool during maintenance, or a function to decide per run — which is how you build permissions:

```

from dataclasses import dataclass
from agents import RunContextWrapper

@dataclass
class UserContext:
    user_id: str
    subscription_tier: str = "free"

def premium_only(ctx: RunContextWrapper[UserContext], agent: Agent) -> bool:
    return ctx.context.subscription_tier in ("premium", "enterprise")

@function_tool(is_enabled=premium_only)
def deep_research(topic: str) -> str:
    """Run an expensive deep research pass on a topic."""
    return f"Deep research on {topic}: ..."

```

A disabled tool isn't refused — it's never mentioned to the model, so the model can't ask for it and can't apologise for it either. The check reads the same context object you built in Part 7.

Failing where the model can see it

A tool that raises kills the run. A tool that *returns* its error keeps the conversation alive and gives the model something to work with:

```

@function_tool
def divide(a: int, b: int) -> str:
    """Divide two numbers."""
    try:
        return str(a / b)
    except ZeroDivisionError:
        return "Error: cannot divide by zero. Ask the user for a different number."

```

The model reads that sentence and asks a better question. For heavier needs — logging, retries, a standard error shape across many tools — `@function_tool(failure_error_function=...)` intercepts exceptions centrally instead.

Recap

Concept	What it gives you
<code>Agent(name, instructions, model, tools)</code>	Defines <i>who</i> the agent is and what it can use
<code>Runner.run_sync / run / run_streamed</code>	Actually executes the agent against an LLM
<code>asyncio</code>	Lets LLM calls (network I/O) run without blocking your whole program
Agent / Run / Global model config	Three scopes for choosing <i>which</i> LLM backs an agent
<code>@function_tool</code>	Turns a Python function into something the agent can call mid-conversation
<code>ModelSettings(temperature, tool_choice, max_tokens)</code>	Tunes <i>how</i> the chosen model behaves on a given run
<code>RunContextWrapper + context=</code>	Passes your own data to tools without ever showing it to the LLM
<code>instructions=<callable></code>	Rebuilds the system prompt on every turn, from context or state
<code>agent.clone(...)</code>	Produces a variant of an agent, overriding only the fields you name
<code>trace(...)</code> / traces dashboard	Records every step of a run so you can inspect it after the fact
<code>agent.as_tool(...)</code>	Turns a specialist agent into a tool another agent can call
<code>handoffs=[...] + result.last_agent</code>	Transfers the conversation to a specialist and tells you who answered
<code>tool_use_behavior, max_turns, is_enabled</code>	Control over when the loop stops and which tools exist

You now have everything needed to build the **First Coded Agent** project: an agent, backed by a provider of your choice, that can reason about a request and reach for a tool when it needs one.

Part 14 — Practice This with Claude Code

Reading code is not the same as building it. This course's approach is to **learn by prompting Claude Code** — Anthropic's agentic coding CLI — and reviewing what it produces, rather than typing everything from scratch or copy-pasting blindly. Open a terminal in your project folder, run `claude`, and work through the prompts below **in order**. Each one builds on the file Claude Code just created or edited for you.

Don't just accept the output — read the diff, ask "why did you do it that way?", and only move to the next prompt once you understand the code you now have.

Prompt-by-part

Part	Prompt to give Claude Code
0. Setup	Scaffold a uv-managed Python project called hello-agent with openai-agents and python-dotenv as dependencies. Create a .env file with empty OPENAI_API_KEY and GEMINI_API_KEY entries, and make sure .env is gitignored.
1. OpenAI key	Create main.py using the OpenAI Agents SDK: an Agent named Assistant with instructions "You are a helpful assistant", run synchronously against the prompt "Write a haiku about recursion in programming." Load OPENAI_API_KEY from .env. Then explain what Agent and Runner.run_sync each do.
2. Gemini key	Modify main.py so the same Agent uses Gemini's gemini-2.5-flash model instead of OpenAI's default – wire up AsyncOpenAI with Gemini's OpenAI-compatible base_url and wrap it in OpenAIChatCompletionsModel. Explain line-by-line why swapping the provider didn't require touching Agent or Runner.
3. Runner & asyncio	Rewrite main.py to call Runner.run inside an async def main() instead of run_sync, executed via asyncio.run(). Then add a second version that fires two different prompts at the same agent concurrently with asyncio.gather and prints both results. Explain why the concurrent version is faster.
4. Model configuration	Refactor my agent so the Gemini model is set at the Run level via RunConfig instead of the Agent level. Show me the same agent being called once with the Gemini run_config and once without one, so I can see the difference. Explain when I'd pick Run-level over Agent-level.
5. Tool use	Add a @function_tool called get_weather(city: str) -> str returning a hardcoded string, and a second tool add(a: int, b: int) -> int. Wire both into the agent's tools list and run it against "What's 42 plus 58, and what's the weather in Lahore?" Then walk me through, step by step, what happened between my prompt and the final answer, including which tools got called and why.
6. Model settings	Take my weather-and-add agent and run the same question three times with model_settings=ModelSettings(tool_choice="auto"), then "required", then "none". Print all three answers side by side and explain what changed and why the "none" answer is different.
7. Local context	Add a UserInfo dataclass with name and uid, pass it to Runner.run via context=, and add a tool that reads wrapper.context.name. Then print the tool's params_json_schema and explain why the wrapper parameter isn't in it.
8. Dynamic instructions	Replace my agent's instructions string with a function taking (context, agent) that greets the user by name from context and mentions the agent's own name. Print the resolved system prompt before the model call, then convert the function into a callable class that counts interactions.

Part	Prompt to give Claude Code
9. Agent cloning	Create a base agent with temperature 0.7 and one tool, then clone it into a creative variant and a precise variant. Prove to me with is-comparisons which attributes are shared between base and clone and which are not.
10. Tracing	Remove set_tracing_disabled from my project, call set_tracing_export_api_key with my OpenAI key, and wrap two consecutive Runner.run calls in a single with trace("...") block. Then tell me exactly which spans I should expect to see on platform.openai.com/traces and in what order.
11. Agents as tools	Build a Writing Coach agent that owns two specialists wrapped with as_tool: a Spanish translator and a summarizer. Run all three of my requests through it, then show me which tool fired for each and explain why the final answer still comes from the coach.
12. Handoffs	Convert my Writing Coach's two as_tool specialists into real handoffs instead. Run the same three requests, print result.last_agent for each, and show me the HandoffCallItem and HandoffOutputItem in result.new_items. Then tell me which of the two designs I should keep and why.
13. Advanced tool control	Add StopAtTools to my agent so the loop ends on a finalizing tool, wrap a run in try/except MaxTurnsExceeded with max_turns=2, and gate one tool behind is_enabled reading a subscription tier from my context. Run it once as a free user and once as premium and show me the difference in the tools the model was offered.

Debug-it prompts (learn by breaking things)

Understanding deepens when you have to *fix* something, not just read it. Try these after finishing Part 5:

- Deliberately break my Gemini setup by using the wrong base_url, show me the exact error it raises, then walk me through how you diagnosed and fixed it.
- Rename the get_weather function to fetch_weather but leave the docstring and tools list referencing the old name. Run it, show me what breaks, then fix it and explain why the SDK cares about the function name at all.
- Remove the type hint from the city parameter in get_weather. Run the agent again – does it still work? Explain what the SDK does differently when a tool parameter has no type hint.
- Change my dynamic instructions function so it takes only the context parameter, run it, and show me the exact TypeError the SDK raises before fixing it back.
- Give my two as_tool specialists the same tool_name, run the orchestrator, and show me what happens. Then fix it and explain how the model tells the two tools apart.
- Rename my Fitness Coach agent to Running Coach but leave the router's instructions mentioning transfer_to_fitness_coach. Run it, show me how the routing degrades, and explain where that tool name comes from.

- Take my local-context example and remove context=user_info from the Runner.run call, leaving the tool untouched. Run it, show me the exact error, and explain which line of the tool blew up and why.

Capstone prompt

Once you've been through all fourteen parts, try building something new end-to-end in one shot, then critique what comes back:

```
Using the OpenAI Agents SDK, build a small CLI agent powered by Gemini (Agent-level configuration) with two tools of your own choosing (not get_weather or add). Expose it through an async main() run via asyncio.run(). Then give me three test prompts that demonstrate the agent choosing between the two tools correctly, and explain the tool-selection reasoning for each one.
```

Compare what Claude Code builds against Parts 1–13 of this guide — if you can explain every line without asking "why," you've learned the material.